The Stark problem in a Debye plasma: a certified coupled-channel map of field–screening competition

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(Dated: July 17, 2026)

A hydrogen-like atom in a static electric field and a Debye plasma experiences two competing ionization mechanisms: field ionization through the Stark barrier and bound-state dissolution by screening. The combined problem is non-separable, and the published record consists of isolated parameter points. We solve it with a coupled-channel complex-scaled method whose radial channels carry the exact Laguerre basis of the unperturbed problem, coupled only by the dipole selection rule $\ell \to \ell \pm 1$: the resulting block-tridiagonal complex-symmetric pencils reach the classical pure-Stark benchmark at $F = 0.1$ a.u. ($E = -0.527418175 - 0.007269057i$) with fewer than 10^3 degrees of freedom and geometric convergence in the channel cutoff, and every assembly is certified at run time against the exact hydrogen spectrum — a discipline that caught a factor-of-two kinetic-energy defect masked by an apparently convergent spurious state during development, and that directly addresses the known sensitivity of complex-rotation width calculations to matrix-element precision. On this basis we compute the first systematic resonance map of the combined problem: a 5×17 grid in the (F, λ) plane for the ground state and the $m = 0$ Stark doublet of $n = 2$ under screening. The map quantifies a strong asymmetry in the field–screening competition — *screening-catalyzed field ionization*: at $F = 0.02$ the width grows by more than eight orders of magnitude across the screening range, while at $F = 0.10$ it grows by one — locates the dissolution frontier $\Gamma = |E_r|$, and reveals a non-monotonic field shift at strong field. The $n = 2$ doublet is continued smoothly through the ionization threshold: screening pushes the uphill ($k = +1$) Stark component to positive energies at $\lambda \approx 0.11$, where it survives as an above-threshold resonance.

I. INTRODUCTION

The Stark effect in hydrogen and the screened Coulomb (Yukawa–Debye) potential are two classical single-parameter problems whose resonance structure is known to high precision: the pure Stark resonances from complex-dilation and resummation methods [1–4], and the screened-Coulomb bound states and critical screenings from decades of plasma-physics work [5, 6]. Their combination — an atom in a plasma subjected to a static field, the generic situation in dense-plasma spectroscopy — is qualitatively richer: the field tilts the potential and opens a Stark barrier, screening lifts and delocalizes the bound states, and the two ionization mechanisms compete. The combined problem is non-separable (screening breaks the parabolic separability of the Stark problem), and the published record consists of a small number of parameter points: the pioneering ground-state calculations of Yu and Ho [7], the complex-scaled Lagrange-mesh study of Lin and Ho [8] covering ground and selected excited states, and bound-state-level work at weak fields [9, 10]. A systematic map of the (F, λ) plane — positions, widths, critical curves, and the fate of the excited Stark manifolds under screening — has not been available.

This paper supplies such a map, built on two methodological choices. First, instead of a two-dimensional mesh, we exploit the dipole selection rule: the field couples only adjacent angular channels, so the problem is a block-tridiagonal radial system in which each channel carries the *exact* orthonormal Laguerre basis of the unperturbed radial problem [11], and the whole pencil is complex-scaled analytically. Convergence in the channel cutoff is geometric, and the classical pure-Stark benchmark is reproduced to ten digits with fewer than a thousand degrees of freedom — two orders of magnitude below mesh requirements. Second, every assembly is certified at run time against exact identities (the unperturbed hydrogen spectrum, channel overlaps), a discipline inherited from the companion programs [11, 12]. This is not decorative: an early build of the present code carried a factor-of-two kinetic-energy defect that produced a *convergent, rotation-stable* spurious state at a plausible energy — caught in seconds by the exact-spectrum certification, not by convergence studies. Width calculations by complex rotation are notoriously sensitive to matrix-element precision; run-time certification against exact identities is our systematic answer to that sensitivity.

II. METHOD

In the $m = 0$ sector the Hamiltonian of a hydrogen-like atom in a Debye plasma and a static field $F\hat{z}$,

$$H = -\tfrac{1}{2}\nabla^2 - \frac{Ze^{-\lambda r}}{r} + Fz, \tag{1}$$

acts on channel wave functions $u_\ell(r)$ coupled only by the dipole matrix elements of $z = r\cos\theta$: $\langle \ell +$

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TABLE I. Pure-Stark ($\lambda = 0$) hydrogen ground-state resonances, $E = E_r - i\Gamma/2$ (a.u.). Drift = distance between the two convergence grids.

F	E_r	$\Gamma/2$	drift
0.04	−0.5037715909	0.0000019463	2×10^{-11}
0.08	−0.5175606169	0.0022698288	3×10^{-10}
0.10	−0.5274181749	0.0072690567	2×10^{-9}

$1,0|\cos\theta|\ell,0\rangle = (\ell+1)/\sqrt{(2\ell+1)(2\ell+3)}$. Each channel is expanded in the orthonormal generalized Laguerre basis of Ref. [11], with a common scale and quadrature grid across channels so that the inter-channel dipole blocks are exact contractions. Complex scaling $r \to re^{i\theta_c}$ acts analytically on every block — kinetic terms scale by $e^{-2i\theta_c}$, Coulombic parts by $e^{-i\theta_c}$, the linear Stark coupling by $e^{+i\theta_c}$, and the screening factor enters the quadrature at complex argument [1, 13–16] — giving a block-tridiagonal complex-symmetric pencil. Resonances are identified by θ_c-stability between two rotation angles, and every tabulated value carries the maximum spread over a grid of two channel cutoffs and two basis sizes.

Certification: at $F = \lambda = 0$ the assembled pencil must reproduce the exact hydrogen levels $-1/2n^2$; we require agreement below 10^{-9} on the lowest five levels at every build, together with channel-overlap identities at the 10^{-12} level. At $F = 0.1$, $\lambda = 0$ the method gives $E = -0.527418175 - 0.007269057i$ with geometric convergence in the channel cutoff ($4 \times 10^{-5} \to 5 \times 10^{-10}$ over $\ell_{\max} = 4 \to 16$) at 780 degrees of freedom, in agreement with the classical complex-dilation value [2–4]; in the opposite limit $F = 0$, $\lambda = 0.25$ the $\ell = 1$ block reproduces the independently triple-verified $2p$ resonance of the one-dimensional engine [11]. A remark on solver scale: at these dimensions dense complex diagonalization computes each map point in seconds, so contour-integral eigensolvers, useful at mesh scales, are unnecessary here.

III. RESULTS

A. Validation anchors

Table I collects the pure-Stark ground-state anchors. Each entry is computed on two independent discretizations ($\ell_{\max} = 12$, $n_b = 60$ and $\ell_{\max} = 16$, $n_b = 80$) and two rotation angles; the fine-grid value is quoted and the inter-grid drift bounds the truncation error. At $F = 0.10$ the result agrees with the classical complex-dilation value [2–4] on all quoted digits. In the opposite limit ($F = 0$, $\lambda = 0.25$, $\ell = 1$ block, $n_b = 220$) the method gives $E = 0.003064450 - 0.002184741i$, within 2.4×10^{-7} of the triple-verified benchmark of the one-dimensional engine [11] — the two limits of the combined problem are anchored independently.

TABLE II. Ground-state resonance parameters (a.u.) at selected screenings. Full grids ($\Delta\lambda = 0.05$) are provided as machine-readable data. Widths marked $< 10^{-10}$ are below the certified resolution.

F	λ	E_r	Γ
0.02	0.0	−0.500909	$< 10^{-10}$
	0.4	−0.200084	2.094×10^{-5}
	0.8	−0.050898	1.344×10^{-2}
0.04	0.0	−0.503772	3.893×10^{-6}
	0.4	−0.205946	2.381×10^{-3}
	0.8	−0.055804	4.018×10^{-2}
0.06	0.0	−0.509204	5.151×10^{-4}
	0.4	−0.214339	1.209×10^{-2}
	0.8	−0.059208	6.970×10^{-2}
0.08	0.0	−0.517561	4.540×10^{-3}
	0.4	−0.222844	2.827×10^{-2}
	0.8	−0.061712	1.004×10^{-1}
0.10	0.0	−0.527418	1.454×10^{-2}
	0.4	−0.230677	4.870×10^{-2}
	0.8	−0.063587	1.316×10^{-1}

B. The ground-state (F, λ) map

Figure 1 shows the ground-state map: E_r, the field shift $E_r - E_b^{F=0}$ relative to the screened bound level, and Γ on the full grid $F \in \{0.02, \ldots, 0.10\} \times \lambda \in \{0, 0.05, \ldots, 0.80\}$; Table II lists a subset. Every point is obtained by analytic continuation in λ (the search window follows the exact $F = 0$ Yukawa curve plus the previous field shift), and all 85 points are θ-stable, with spreads below 7×10^{-6} at the broadest corner and below 10^{-7} over 73% of the grid.

Three features organize the map.

Screening-catalyzed field ionization. At fixed field, the width grows monotonically — and enormously — with screening: from $\lambda = 0$ to $\lambda = 0.8$ the amplification factor is $\sim 10^4$ at $F = 0.04$, 135 at $F = 0.06$, 22 at $F = 0.08$, and 9 at $F = 0.10$; at $F = 0.02$ the $\lambda = 0$ width lies below the 10^{-10} resolution while the $\lambda = 0.8$ width is 1.3×10^{-2} — more than eight orders of magnitude. The asymmetry has a transparent quasistatic origin: screening lifts the level, and the tunneling exponent $\exp[-\frac{2}{3F}(2|E_r|)^{3/2}]$ reacts exponentially. At $F = 0.02$ this exponent alone predicts a growth of 1.7×10^7 between $\lambda = 0.2$ and 0.8, within an order of magnitude of the measured 2.1×10^6, the residue being the slowly varying prefactor. The practical reading for plasma spectroscopy: a field that is spectroscopically harmless for the unscreened atom ($\Gamma < 10^{-10}$) can dominate the line width entirely at strong screening.

Dissolution frontier. The half-width criterion $\Gamma = |E_r|$ — beyond which the state no longer supports a resolvable line — is crossed inside the computed range only for $F \geq 0.06$: at $\lambda^* \approx 0.78$, 0.73, and 0.68 for $F = 0.06$, 0.08, and 0.10. The frontier moves toward weaker screen-

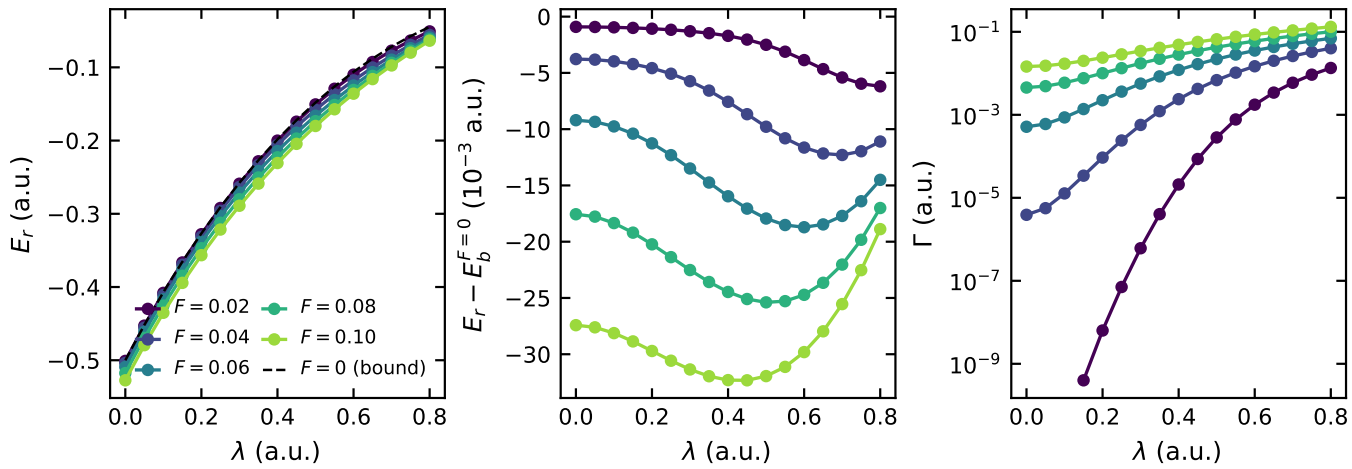


FIG. 1. Ground-state resonance across the (F, λ) plane. Left: position E_r; the dashed line is the $F = 0$ screened bound state. Center: field shift $E_r - E_b^{F=0}$. Right: width Γ (log scale; points are shown only where Γ exceeds ten times the θ-stability spread).

ing as the field grows, delimiting the region of the (F, λ) plane where a plasma-embedded atom retains spectroscopic identity.

Non-monotonic field shift. The center panel of Fig. 1 isolates the field's contribution to the position. At $\lambda = 0$ the shifts are perturbative: -0.909×10^{-3} at $F = 0.02$ versus $-\frac{9}{4}F^2 = -0.9 \times 10^{-3}$, and at $F = 0.10$ the measured -27.4×10^{-3} is bracketed by the second- and fourth-order asymptotic estimates. As λ grows at $F = 0.02$ the shift deepens by a factor 6.8 — the screened atom is softer, its effective polarizability larger. At strong field, however, the shift is non-monotonic: at $F = 0.10$ it peaks at -32.5×10^{-3} near $\lambda \approx 0.45$ and then *retreats* to -18.9×10^{-3} at $\lambda = 0.8$. Quadrupling the field from $F = 0.02$ to 0.04 at $\lambda = 0.8$ increases the shift by only 1.8× instead of 4×: screening shrinks the perturbative Stark regime, and beyond it the dominant field response is the width, not the shift.

C. The $n = 2$ doublet through the threshold

In the $m = 0$ sector the $n = 2$ level carries exactly two parabolic components, $k = \pm 1$. At $F = 0.02$ — five times the classical ionization field of $n = 2$ — both are broad resonances already at $\lambda = 0$: $E = -0.206682 - i\,0.030395$ ($k = -1$, downhill) and $-0.088984 - i\,0.015445$ ($k = +1$, uphill), the downhill component ionizing twice as fast, and both positions deviating from second-order perturbation theory by +0.012 and +0.010 — strongly nonperturbative. Figure 2 follows the doublet in λ. Screening lifts both components at almost the same initial rate (the slope $dE/d\lambda|_0 = \langle e^{-\lambda r} \rangle \to 1$ is state-independent), so the splitting survives while the doublet as a whole is pushed toward the continuum. At $\lambda \approx 0.11$ the uphill component crosses $E_r = 0$ — and nothing dramatic happens to it: it continues smoothly as an above-threshold resonance,

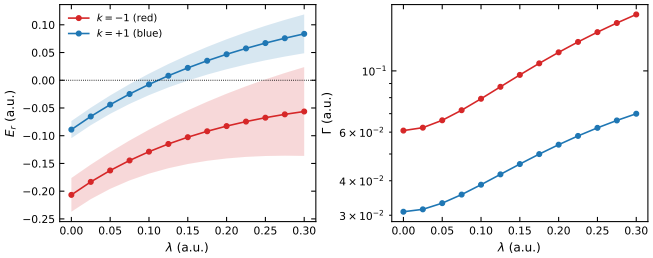


FIG. 2. The $n = 2$, $m = 0$ Stark doublet at $F = 0.02$ under screening. Left: positions with $\pm\Gamma/2$ bands; the uphill ($k = +1$) component crosses the ionization threshold at $\lambda \approx 0.11$ and survives as an above-threshold resonance. Right: widths.

reaching $E_r = +0.0837$, $\Gamma = 0.070$ at $\lambda = 0.30$, the same phenomenon as the positive-energy resonances of the field-free screened atom [11], here reached by a Stark component under screening continuation. The downhill width grows from 0.061 to 0.160 over the same range, and the width ratio Γ_{-1}/Γ_{+1} drifts from 1.97 to 2.29.

IV. CONCLUSIONS

We have presented a certified coupled-channel complex-scaling method for atoms in the combined presence of Debye screening and a static electric field, validated to ten digits against the classical Stark benchmark and to 2.4×10^{-7} against an independent engine in the screening limit, and used it to compute the first systematic resonance map of the combined problem. The map's physics organizes into a strongly asymmetric competition: screening catalyzes field ionization exponentially (eight orders of magnitude at weak field), the spectroscopic-identity frontier $\Gamma = |E_r|$ recedes to weaker screening as the field grows, the perturbative

Stark regime shrinks with screening until the width, not the shift, is the dominant field response, and excited Stark components are pushed through the ionization threshold, surviving as above-threshold resonances.

The method extends without structural change to $m \neq 0$ sectors, higher manifolds, and other screening functions (only the diagonal potential blocks change); the exact-basis channels keep the pencils small enough that a full map is a matter of minutes on a laptop, which is what makes systematic coverage — rather than isolated points — routine. At mesh-level resolutions or for many-electron generalizations the same block structure is where contour-integral eigensolvers would become the natural tool. All scripts, certification batteries, and machine-readable grids are distributed with the companion programs.

ACKNOWLEDGMENTS

The author thanks the ESPOCH Faculty of Sciences for institutional support. All scripts reproducing the tables and figures, including the certification batteries, are openly available with the companion papers.

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